

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	05 July 2026
Team ID	
Project Name	House Rent
Maximum Marks	4 Marks

Technical Architecture:

Use the architecture diagram that we already created (React → Express.js → MongoDB). Since your project does not use any third-party APIs, cloud services, or machine learning, do not include those blocks.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface (Web Application)	Responsive web interface for Admin, Owner, and Renter to interact with the system.	HTML5, CSS3, JavaScript, React.js, Bootstrap
2.	Application Logic-1	Frontend logic for user interaction, navigation, and UI rendering.	React.js
3.	Application Logic-2	Backend logic for authentication, property management, booking, and transactions.	Node.js, Express.js
4.	Application Logic-3	User authentication and password encryption.	JWT (JSON Web Token), bcrypt.js
5.	Database	Stores user details, properties, bookings.	MongoDB
6.	Cloud Database	Not Applicable (Project uses local MongoDB database).	Not Applicable
7.	File Storage	Stores uploaded property images on the local system.	Local File System
8.	External API-1	Not Applicable	Not Applicable

9.	External API-2	Not Applicable	Not Applicable
10.	Machine Learning Model	Not Applicable	Not Applicable
11.	Infrastructure (Server / Cloud)	Application deployed on local system using Node.js server and MongoDB database. Local Server Configuration: Node.js, Express.js, MongoDB Cloud Server Configuration: Not Applicable	Local Server.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The application is developed using open-source frontend and backend frameworks.	React.js, Node.js, Express.js, Bootstrap
2.	Security Implementations	Implements secure user authentication, password encryption, and role-based access control for Admin, Owner, and Renter.	JWT, bcrypt.js, Express Middleware
3.	Scalable Architecture	Designed using a three-tier architecture separating presentation, business logic, and database layers for easy scalability.	React.js, Express.js, MongoDB
4.	Availability	Designed using a three-tier architecture separating presentation, business logic, and database layers for easy scalability.	Node.js Server, MongoDB
5.	Performance	Optimized React components, efficient MongoDB queries, and responsive UI ensure fast loading and smooth user experience.	React.js, MongoDB, Express.js